

Group-Delay Equalization for Complex IIR Filters: Design Procedure

This section states the complete equalizer design procedure on its own, in a form intended to be reusable close to verbatim in a paper.

1 Filter model and all-pass building block

Let $H(z)$ be a (possibly complex, non-conjugate-symmetric) discrete-time IIR filter with passband $[f_{p1}, f_{p2}]$ in normalized cyclic frequency ($F_s = 1$), and let $\tau_H(f)$ denote its group delay. Complex IIR filters designed without an arithmetic-symmetry constraint typically exhibit a markedly non-flat, “bathtub”-shaped group delay: two peaks near the passband edges bridged by a lower floor across the interior. The goal is a cascaded all-pass equalizer that flattens τ_H across the passband without altering $|H(e^{j\omega})|$.

The equalizer is a cascade of N first-order discrete all-pass sections. Each section is parameterized by a single complex pole $p_i = r_i e^{j\theta_i}$, $0 < r_i < 1$:

$$A_i(z) = \frac{z^{-1} - \bar{p}_i}{1 - p_i z^{-1}}. \quad (1)$$

This construction is magnitude-preserving for any complex p_i : $|A_i(e^{j\omega})| = 1$ for all real ω , so cascading A_i onto H leaves $|H(e^{j\omega})|$ unchanged and only modifies phase/group delay. The group-delay contribution of a single section is the Poisson kernel

$$\tau_i(\omega) = \frac{1 - r_i^2}{1 - 2r_i \cos(\omega - \theta_i) + r_i^2}, \quad \tau_i(\omega) > 0 \quad \forall \omega. \quad (2)$$

Because $\tau_i(\omega) > 0$ everywhere, an all-pass cascade can only *add* delay: it can raise a low floor toward the level of existing peaks, but it cannot directly lower a peak, and a section placed too close to an existing peak only adds to it.

For implementation convenience, sections are grouped into M *clusters*: cluster i consists of c_i identical stacked sections sharing one pole (r_i, θ_i) , so $N = \sum_i c_i$, and the total equalizer delay is

$$\tau_{eq}(\omega; \boldsymbol{\theta}, \mathbf{r}) = \sum_{i=1}^M c_i \tau_i(\omega; r_i, \theta_i), \quad \tau(\omega) = \tau_H(\omega) + \tau_{eq}(\omega). \quad (3)$$

2 Design objective: anchored, gain-weighted equi-ripple balance

Fix an **anchor** level once, from the *unequalized* filter alone:

$$D_0 = \text{mean} \left(\tau_H(f) : f \text{ a local extremum of } \tau_H \right), \quad (4)$$

in practice the mean of the filter’s own two edge-of-passband peaks. D_0 is never recomputed once fixed, which keeps the objective anchored to an external, fixed target rather than to the equalizer’s own (evolving) output – an unanchored objective was found in practice to let the optimizer invent new peaks and match them to each other at an arbitrarily poor common level.

For a given $(\boldsymbol{\theta}, \mathbf{r})$, let $\{f_k\}_{k=1}^K$ be the frequencies of *every* local extremum (maxima and minima) of $\tau(\cdot)$ over the passband expanded by a margin fraction μ on each side (default $\mu = 0.10$), redetected from scratch at every iteration (Section 3). Define the gain-weighted deviation from the anchor at each extremum:

$$w_k = |\tau(f_k) - D_0| \cdot |H(f_k)|, \quad k = 1, \dots, K. \quad (5)$$

The design goal is the classical Chebyshev equi-ripple condition: choose $(\boldsymbol{\theta}, \mathbf{r})$ so that all w_k are equal. Weighting by the filter’s own gain $|H(f_k)|$ allows an extremum sitting where $|H|$ has already rolled off to carry a proportionally larger raw delay deviation without violating equality of the weighted terms – the deviation that matters is the one seen at the filter’s output, not the raw group-delay number.

3 One Newton/least-squares iteration

Given a current configuration $(\boldsymbol{\theta}, \mathbf{r})$, one iteration proceeds in two steps.

Step 1 – extremum detection and refinement. Evaluate $\tau''(\omega)$ on a uniform grid over the expanded band and locate its zero crossings (each pair of consecutive crossings, plus the two boundary segments, brackets at most one true extremum of τ , since τ' is monotonic wherever τ'' keeps one sign). Within each occupied bracket, confirm the extremum type from the sign of τ' at the bracket’s endpoints, then refine its location with one Newton step:

$$f_k \leftarrow f_k - \frac{\tau'(f_k)}{\tau''(f_k)}. \quad (6)$$

Step 2 – parameter update. Because cluster i ’s contribution depends only on its own (r_i, θ_i) , the Jacobian of τ with respect to each cluster parameter is available in closed form:

$$\frac{\partial \tau(\omega)}{\partial \theta_i} = c_i \frac{2r_i(1 - r_i^2) \sin(\omega - \theta_i)}{D_i(\omega)^2}, \quad (7)$$

$$\frac{\partial \tau(\omega)}{\partial r_i} = c_i \left(\frac{-2r_i}{D_i(\omega)} - \frac{2(1 - r_i^2)(r_i - \cos(\omega - \theta_i))}{D_i(\omega)^2} \right), \quad (8)$$

$$D_i(\omega) = 1 - 2r_i \cos(\omega - \theta_i) + r_i^2. \quad (9)$$

Assemble the Jacobian $J \in \mathbb{R}^{K \times M}$ (angle-only) or $J \in \mathbb{R}^{K \times 2M}$ (joint angle+radius) of the weighted deviations, row k given by $J_{k,i} = \text{sign}(\tau(f_k) - D_0) |H(f_k)| \partial \tau(f_k) / \partial \theta_i$ (and analogously for $\partial / \partial r_i$). Solve, via the Moore–Penrose pseudoinverse (the system is neither reliably square nor consistently over/under-determined, since K changes call to call),

$$\Delta = J^+(\bar{w} \mathbf{1} - \mathbf{w}), \quad \bar{w} = \frac{1}{K} \sum_{k=1}^K w_k, \quad (10)$$

the least-squares parameter step that drives every w_k toward their common mean, then apply a damped update

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \Delta_{\boldsymbol{\theta}}, \quad \mathbf{r} \leftarrow \mathbf{r} + \alpha \Delta_r \quad (\text{if radii are free}), \quad (11)$$

with damping factor $\alpha \in (0, 1]$, and radii clamped to $[r_{\min}, r_{\max}] = [0.3, 0.995]$ after the update.

4 Two-phase iteration schedule

Phase 1 (angle-only). Radii held fixed; repeat Section 3 with damping $\alpha = \alpha_1$ until the ripple spread $S = \max_k w_k - \min_k w_k$ changes by less than a tolerance ϵ between consecutive iterations, or a maximum iteration budget is reached. This phase converges monotonically in practice.

Phase 2 (joint angle + radius). Continuing from the Phase 1 result, free the radii and repeat Section 3 with damping $\alpha = \alpha_2$, now updating J 's r -columns as well. This phase does *not* converge monotonically: S reaches a minimum after a small number of steps, then drifts upward again while flatness just outside the nominal passband continues to improve – a genuine trade-off, not simple convergence to a fixed point. Accordingly, Phase 2 tracks the best (smallest) S seen so far and stops after P consecutive non-improving iterations, **returning the best configuration found**, not the final iterate.

5 Starting configuration and step-size selection

The iteration is initialized with M clusters of c stacked sections each, pole angles spread across the passband (evenly, or symmetrically if the filter's own specification is symmetric), and a common starting radius r_0 . Two empirical rules govern the free parameters:

- **Damping must shrink as M grows relative to K .** With more free angle/radius parameters against roughly the same number of tracked extrema, the pseudoinverse step becomes more aggressive per unit residual; the default $\alpha = 0.5$, stable at $M = 3$, produces outright divergence at $M = 5$ (oscillating extremum count, radii pinned at their clamp bounds) unless α_1, α_2 are reduced (e.g. to 0.2/0.1, or smaller still when perturbing a starting point that is already known to be reasonable).
- **Lower r_0 suits wider passbands.** A single section's own peak height $(1 + r_i)/(1 - r_i)$ grows far faster with r_i than its fractional leakage to distant frequencies shrinks; for a passband spanning a large fraction of the full cycle, a lower, broader starting radius blends clusters into a smooth correction rather than a sequence of narrow, high- Q bumps.

6 Algorithm summary

Algorithm 1 Anchored equi-ripple group-delay equalization

Require: $H(z)$; passband $[f_{p1}, f_{p2}]$; expansion margin μ ; cluster count M ; stages per cluster c_i ; starting angles $\boldsymbol{\theta}^{(0)}$ and radius r_0 ; damping α_1, α_2 ; tolerance ϵ ; patience P .

- 1: Compute τ_H over the expanded band; set anchor D_0 from τ_H 's own extrema (Eq. 4).
- 2: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}^{(0)}$; $\mathbf{r} \leftarrow r_0 \mathbf{1}$.

Phase 1 (angle-only):

3: **repeat**

4: Update $\boldsymbol{\theta}$ via Section 3 with radii fixed, damping α_1 .

5: **until** $|S^{(t)} - S^{(t-1)}| < \epsilon$

Phase 2 (joint angle+radius):

6: $S_{\text{best}} \leftarrow S$; $(\boldsymbol{\theta}_{\text{best}}, \mathbf{r}_{\text{best}}) \leftarrow (\boldsymbol{\theta}, \mathbf{r})$; $\text{noImprove} \leftarrow 0$

7: **repeat**

8: Update $(\boldsymbol{\theta}, \mathbf{r})$ via Section 3 with radii free, damping α_2 .

9: **if** $S < S_{\text{best}}$ **then**

10: $S_{\text{best}} \leftarrow S$; $(\boldsymbol{\theta}_{\text{best}}, \mathbf{r}_{\text{best}}) \leftarrow (\boldsymbol{\theta}, \mathbf{r})$; $\text{noImprove} \leftarrow 0$

11: **else**

12: $\text{noImprove} \leftarrow \text{noImprove} + 1$

13: **end if**

14: **until** $\text{noImprove} \geq P$

15: **return** equalizer $\{(r_i, \theta_i, c_i)\}_{i=1}^M = (\boldsymbol{\theta}_{\text{best}}, \mathbf{r}_{\text{best}}, \mathbf{c})$
